

SharpCommunicator

by Martin Erzberger

Use Case Documentation

for the Sharp Pocket Computers PC-1500, PC-1500A and PC-1600
for Serial Communication with the CE-158X Module by Soigerneris (USA)

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Introduction

The SharpCommunicator is a program written by Martin Erzberger. The program can be found and downloaded at github:

```
> https://github.com/tinue/sharp-pocket-computer/releases
```

Always use the most recent version (even if it is a pre-release). This documentation has been written based on the version 2.1.0 from 17th December 2024. The program is written in Java and requires a JDK or JRE (Java Development Kit or Java Runtime Environment, i have open-jdk 21 installed on my Linux Mint 22 Laptop and it works perfect. Java is available for other operating systems like Windows or MacOS, too. So you can install Java and use the SharpCommunicator from those OS's also. The tool does not have a gui – is only available via command-line. I use a bash shell environment on my laptop. Probably i (or someone else is going to create some shell scripts to simplify the useage of the SharpCommunicator but as it needs some time, i currently do not have, someone else likes to start that job.

Of course the CE-158X needs to be connected to the PC-1500 and you should have read and understood the documentation from Jeff for his wonderful little box. Best is you read the documentation of the SharpCommunicator also. However, the documentation is currently not completely up to date. This is opensource and created in the spare time of a nice man – so nothing to complain about – just my opinion. By the way – the documentation is „hidden“ in the zip files with the source code of the SNAPSHOT or other release of Martin's SharpCommunicator.

If you are going to transfer ASCII Basic files to the PC-1500, make sure the line ending is done in the MacOS way, simply with a 0x0D. This is very important, as otherwise you get errors and the data is not transferred.

The 'use cases' documented here assume that the interface on the CE-158x (USB) has been initialised as follows:

```
> SETCOM 19200,8,N,1
```

```
> SETDEV U1, CI,C0
```

If you are using the U0 interface or even the original CE-158 from Sharp, the second command needs to look as follows (if U1 is not mentioned, U0 is default):

```
> SETDEV CI,C0
```

1. Transfer ASCII Basic File from PC to PC-1500

On the Sharp PC-1500 just type

```
> CLOAD <ENTER>
```

The command line used to transfer the file „disass.bas“ from my PC (Linux Mint 22) to my PC-1500 using a CE-158X is:

```
> java -jar SharpCommunicator.jar -d pc1500 -i disass.bas
```

Make sure to start the CLOAD command first and only then the SharpCommunicator!

What happens: the SharpCommunicator transforms the ASCII program file „disass.bas“ into a tokenized version and transfers it to the PC-1500. The parameter „-d pc1500“ is not necessary, as the pc1500 is the default in that process. If you want to transfer to the PC-1600 then you need the parameter „-d pc1600“.

Transfer takes about 10 seconds with the program „disass.bas“ which has a size of 8,2KB.

2. Transfer a Binary Program from PC to PC-1500

On the Sharp PC-1500 just type

```
> CLOAD <ENTER>
```

The command line used to transfer the just the binary version of the „disass.bas“, „disassem“ also from the PC (same as above) to my PC-1500 is:

```
> java -jar SharpCommunicator.jar -d pc1500 -i disassem
```

Make sure to start the CLOAD command first and only then the SharpCommunicator!

What happens: the SharpCommunicator recognizes the „disassem“ file to be a binary file (already tokenized) and sends it to the Sharp PC-1500.

3. Transfer a Binary Program from PC-1500 to PC

The command line used to transfer the just transferred „disass.bas“ back from the PC-1500 to my PC (same as above) is:

```
> java -jar SharpCommunicator.jar -d pc1500 -o disassem
```

On the Sharp just type:

```
> CSAVE <ENTER>
```

Make sure to start the SharpCommunicator first and only then start the CSAVE command on the PC-1500!

What happens: the CSAVE command sends the program in tokenized form to the PC. Likely the „-d pc1500“ is obsolete, i leave it here to point out that you would need a „-d pc-1600“ if you are working with a PC-1600. About 3-4 seconds after the transfer has completed, the

SharpCommunicator returns back to the command line and the file after the -o parameter is on your PC.

If for whatever reason the SharpCommunicator doesn't recognize transfer-end (give it 5 seconds), you might just press CTRL-C in order to stop the SharpCommunicator. Should do the same job.

4. Transfer an ASCII Program from PC-1500 to PC

The command line to transfer an ASCII Basic File from the PC-1500 to the PC is:

```
> java -jar SharpCommunicator.jar -d pc1500 -o disassem.asc
```

On the Sharp PC-1500 just type:

```
> CSAVEa <ENTER>
```

Make sure to start the SharpCommunicator first and only then the CSAVEa command on the PC-1500.

The process is very slow compared to the transfer of binary programs/data, so be patient. Some seconds after the transfer is completed from the PC-1500 CSAVEa command, the SharpCommunicator recognizes that there is no more traffic and returns to command line. I rare cases you might have to press CTRL-C to complete the job.

5. Transfer a Machine Language Programm from PC-1500 to PC

The command line used to transfer a machine language program from the PC-1500 to my PC (same as above) is:

```
> java -jar SharpCommunicator.jar -d pc1500 -o test.bin
```

On the Sharp just type:

```
> CSAVEM „test.bin“;&start-addr,&end-addr[,exec-addr]<ENTER>
```

Make sure to start the SharpCommunicator first and only then start the CSAVEM command on the PC-1500! And regard the capital M. All other CLOAD/CSAVE commands do use a lowercase command extension (r for Reserve, a for ASCII).

6. Transfer a Machine Language Programm from PC to PC-1500

On the Sharp PC-1500 just type

```
> CLOADM <ENTER>
```

Or, if you want to relocate the loaded program/data to a different memory address

```
> CLOADM „test.bin“;&new-addr<ENTER>
```

The command line used to transfer the the machine language program to the PC-1500 is:

```
> java -jar SharpCommunicator.jar -d pc1500 -i test.bin
```

Make sure to start the CLOADM command first and only then the SharpCommunicator!

What happens: the SharpCommunicator recognizes the incoming file to be a binary file and sends it to the Sharp PC-1500. The Sharp PC-1500 itself expects binary data as it has launched the CLOADM command (>M for Machine Language). And regard the capital M. All other CLOAD/CSAVE commands do use a lowercase command extension (r for Reserve, a for ASCII).

7. Transfer a Reserve Memory Snapshot from PC-1500 to PC

The command line used to transfer the reserve memory content from the PC-1500 to my PC (same as above) is:

```
> java -jar SharpCommunicator.jar -d pc1500 -o reserve
```

On the Sharp just type:

```
> CSAVER <ENTER>
```

Make sure to start the SharpCommunicator first and only then start the CSAVER command on the PC-1500!

What happens: the CSAVER command sends the data from the reserve memory area to the PC. Likely the „-d pc1500“ is obsolete, i leave it here to point out that you would need a „-d pc-1600“ if you are working with a PC-1600.

Currently there is one flaw in the process (same as with sending binary programs). When the PC-1500 has sent the whole reserve memory, the SharpCommunicator does not recognize the end of file and still waits. If you are sure everything is transferred - just give the process some seconds to complete after CSAVER has stopped on the PC-1500 – just press CTRL-C to stop the process manually. The file is already written and you now have a copy of the reserve oin the file „reserve“ on your PC.

8. Transfer a Reserve Memory Snapshot from PC to PC-1500

On the Sharp PC-1500 just type

```
> CLOADr <ENTER>
```

The command line used to transfer the just the binary version of the „disass.bas“, „disassem“ also from the PC (same as above) to my PC-1500is:

```
> java -jar SharpCommunicator.jar -d pc1500 -i reserve
```

Make sure to start the CLOADr command first and only then the SharpCommunicator!

What happens: the SharpCommunicator recognizes the „reserve“ file to be a binary file and sends it to the Sharp PC-1500. The Sharp PC-1500 itself expects binary data as it has launched the CLOADr command (r for reserve).

9. Some Additional Explanations to the Command Line Options

Just for completeness, here the currently available (and working) command line options of SharpCommunicator:

usage: SharpCommunicator

-d, --device <arg>	Device to use (pc1500, pc1500a or pc1600). Default is pc1500.
-h, --help	Print usage.
-i, --in-file <arg>	Input file. If this argument is missing, the input is taken from the PC-1500/1600 via serial.
-if, --in-format <arg>	Format of the input file (ascii or binary). Default is ascii for files with an extension of .bas, binary for all other files.
-o, --out-file <arg>	Output file. If this argument is missing, the output is sent to the PC-1500/1600 via serial.
-of, --out-format <arg>	Format of the output file (ascii, asciicompact or binary). Default is ascii for files with an extension of .bas, binary for all other files.
-ra, --autorun-address <arg>	Autorun address of a machine language program. Pass the value in hex, for example '38c5' or '38C5'. If the parameter is skipped, the program will not start by itself after loading
-sa, --start-address <arg>	Start address of a machine language program. Pass the value in hex, for example '38c5' or '38C5'
-u, --add-utils	Add the serial utilities when loading the Basic program.
-v, --verbose	Verbose mode (more logging).
-V, --version	Prints version information.
-vv, --debug	Debug mode (even more logging).

There are plans to add more options and capabilities to the SharpCommunicator – however, everything needs its time.

9.1. -d / --device [pc1500 | pc1500a | pc1600]

That options gives the PC-1500 or PC-1600 model, if you give no -d option, the PC-1500 is default. So in all above examples the -d pc1500 was obsolete as i have done all the transfers between my Linux laptop and my PC-1500 with CE-158x.

9.2. -h / --help

This is quite self-explanatory – this switch simply shows the help text, which you likely ave already seen, when you mis-typed an option.

9.3. -i,--in-file [filename]

If you want to send a file to the Sharp PC-1500 (or other) this option/switch allows you to specify the filename of that file. If this argument is missing, the input is taken from the PC-1500/1600 via serial. If both -i and -o (see below) are missing, the SharpCommunicator gives an error.

9.4. -o,--out-file [filename]

If you want to receive a file from your Sharp PC-1500 (or other) this option/switch allows you to specify the filename for the file to be saved to disk. If this argument is missing, the output is sent to the PC-1500/1600 via serial. So with -i filename given and no -o, the input file from the laptop is sent to the PC-1500 via serial port, as long as there is no -o argument given. With neither of these two parameters (-i, -o) the SharpCommunicator gives an error message.

9.5. -if,--in-format [ascii | binary]

If you want so send a file to the Sharp PC-1500, the default format for the transfer is ascii for files with an extension .bas and binary for the others. So if you want to override this default, just use this switch. So for example lets assume, you want to transfer a file disassem.asc to the PC-1500, which is in ascii format on the disk of you laptop. To get it trasferred in ascii mode, just use the switch

9.6. -of, --out-format [ascii | asciicompact | binary]

If you want so send a file from the Sharp PC-1500 to your laptop the default format for the transfer is ascii for files with an extension .bas and binary for the others. So if you want to override the default (i.e. for a file named disassem.asc) use this switch.

9.7. -ra, --autorun-address <arg>

Autorun address of a machine language program. Pass the value in hex, for example '38c5' or '38C5'. If the parameter is skipped, the program will not start by itself after loading.

9.8. -sa, --start-address <arg>

Star address of a machine language program. Pass the value in hex, for example '38c5' or '38C5'.

9.9. -u, --add-utils

This option simply adds some lines of code to the transferred program:

```
61000 "J":PRINT "SERIAL INIT"  
61010 SETDEV U1,CI,CO  
61080 END  
61100 "S":PRINT "SAVING"  
61110 CSAVEa  
61120 END  
61200 "L":PRINT "LOADING"  
61210 CLOADa  
61220 END
```

9.10. -v --verbose

Verbose mode results in more logging information.

9.11. -V --version

Simply prints the version information of the currently used release of SharpCommunicator.

9.12. -vv --debug

Debug mode results in much more logging information.

This might change with incrementing releases. For example the CSAVEa and CLOADa might change to CSAVE and CLOAD. Also a SETCOM command might be added.

8. Make it easy!

On Linux and Windows one can make things easier by using scripts. I have made two tiny scripts to make the usage of Sharp Communicator more easy. I have called the scripts „cload“ and „csave“ (surprise, surprise) to indicate the direction on the PC-1500 (or different pocket computer).

8.1 CLOAD

Here comes the CLOAD script. The documentation is only in the script at the moment.

```
#!/bin/bash
#
# to transfer a program from your Linux PC in ASCII format, use CLOAD on the pocket
# the command on Linux bash should then look like "cload <programname>.bas" (rename
# from .asc first)
# if you use .asc extension, SharpCommunicator will give an error message
#
# to transfer a binary basic program (tokenized) from your Linux PC, use simply CLOAD on the pocket
# the command on the Linux bash should then look like "cload <programname>.bin"
# if you use .bas the SharpCommunicator gives an error message
#
# to transfer the content of the reserve memory area from your Linux PC, use CLOADr on the pocket
# the command on the Linux bash should then look like "cload <reserve>.bin"
#
# to transfer a machine language program from your Linux PC use 'CLOADM' unless you
# want to relocate the machine language program, then use
# 'CLOADM "program.bin";&reloc-addr
# the command on the Linux bash should then look like "cload <program>.bin"
#
# change the path according to your needs
sharpcommunicator="/home/xian/Desktop/SharpCommunicator/SharpCommunicator.jar"
# change your pocket device to your needs, too
pocketdevice="pc1500"
#
# -----
#
if [ -z "$1" ]; then
    echo "-----";
    echo "Script requires a filename as an argument";
    echo "-----";
    exit 1;
fi
java -jar $sharpcommunicator -d $pocketdevice -i $1
#
# completed
# -----
```

8.2 CSAVE

Here comes the CSAVE script. The documentation is only in the script at the moment.

```
#!/bin/bash
#
# to transfer the program on your pocket in ASCII format, use CSAVEa on the pocket
# the command on Linux bash should then look like "csave <programname>.asc"
#
# to transfer a binary basic program (tokenized) to your Linux PC, use CSAVE on the pocket
# the command on the Linux bash should then look like "csave <programname>.bin"
# if you use .bas the SharpCommunicator gives an error message
#
# to transfer the content of the reserve memory area to your Linux PC, use CSAVER on the pocket
# the command on the Linux bash should then look like "csave <reserve>.bin"
#
# to transfer a machine language program to your Linux PC use
# 'CSAVEM "program.bin";&from-addr,&tp-addr[,&exec-addr]
# the command on the Linux bash should then look like "csave <program>.bin"
# the from-address, the length (calculated from end-address) and the execution
# address are stored automatically
#
#
# change the path according to your needs
sharpcommunicator="/home/xian/Desktop/SharpCommunicator/SharpCommunicator.jar"
# change your pocket device to your needs, too
# can be pc1500, pc150a, pc1600
pocketdevice="pc1500"
#
# -----
#
#
if [ -z "$1" ]; then
    echo "-----";
    echo "Script requires a filename as an argument";
    echo "-----";
    exit 1;
fi

echo "--> now start the csave-command on your pocket computer"
java -jar $sharpcommunicator -d $pocketdevice -o $1
#
# completed
# -----
```

9. Final Remarks

The SharpCommunicator has some more options to play with and explore. However, this documentation is just for the purpose to document the most important use cases for the software and help getting forward with transferring data between the PC-1500 and the laptop (or whatever PC). I was fiddling around with CoolTerm and not really happy with it. It failed far too often in my opinion and all parameters need to be 100%. Then you need to take care that the parameters for transferring ASCII data are different than the parameters for binary data.

I was very happy that Martin has written the SharpCommunicator and had helped me to understand and use it so that i could do the job i intended to do.

As SharpCommunicator is a work in progress software, also this document will likely grow and be expanded as necessary and possible.

10. Revision History

Version	Date	Content
0.1	2.12.2024	First version; based on software version 1.1.0
0.2	5.12.2024	Update on some points; based on software version 2.0.0 pre-release
0.3	6.12.2024	Major update due to Software changes; based on software verison v2.0.1-prerelease
0.4	8.12.2024	Bug fix, no changes except of version
0.5	9.12.2024	Bug fix, no changes except of version
0.6	14.12.2024	Bug fix, no changes except of version
0.7	14.05.2025	Some further explanations and documentation of two scripts to make it more easy
0.8	31.08.2025	Added new command line options from the latest version 2.1.0 pre prelease. Not yet described.